Forced mixed corners and low-facet rigidity for right-angled hyperbolic polytopes

Research manuscript with reproducible exact calculations

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Abstract

Any finite-volume right-angled hyperbolic polytope in dimension eleven or twelve contains a specific ten-facet mixed four-face. We derive explicit finite/ideal vertex ratios and positive occurrence bounds using a weighted two-face inequality. We also classify equality in the four-dimensional facet-plus-cusp bound fifteen and in the five-dimensional facet bound sixteen. A complete five-dimensional corner forces either that sixteen-facet polytope or at least twenty facets; in every other five-polytope such corners occupy at most one eleventh of the finite vertices. These are universal necessary conditions, with no arithmetic or facet-spectrum restriction. They do not improve the published dimension ceiling twelve. Exact rational calculations, finite model checks and a numerical feasibility diagnostic are supplied; the geometric proofs have not yet been formalized in Lean.

1 Statements and conventions

Let $P \subset \mathbb{H}^n$ be a finite-volume right-angled polytope. Write f, c, a_j for its finite vertices, ideal vertices and j-faces. Faces of positive dimension inherit right angles and finite volume. At finite vertices the links are simplices; at ideal vertices they are rectangular boxes. Outward facet normals have square one, and the Lorentz metric has one negative direction.

Let P_3 be the six-facet triangular bipyramid with two finite and three ideal vertices. Let P_4 be the ten-facet polytope with five finite and five ideal vertices described in Section 3. Let P_5 be the sixteen-facet polytope with sixteen finite and ten ideal vertices described in Section 5. These are the known Potyagailo–Vinberg polytopes [1]. Let T count triangular two-faces with one finite vertex, and let B, C, D count faces isometric to P_3, P_4, P_5, respectively.

Theorem 1. *For $n = 12$,*

$$\frac{1474}{21}c < f < \frac{1298}{9}c, \qquad T > \frac{331}{437}a_2, \qquad T > \frac{1986}{59}f, \qquad C > \frac{627}{1475}f.$$

For $n = 11$,

$$\frac{268}{7}c < f < \frac{236}{3}c, \qquad T > \frac{331}{437}a_2, \qquad T > \frac{1655}{59}f, \qquad C > \frac{1763}{5310}f.$$

In either dimension P has a four-face isometric to P_4.

Theorem 2. *A right-angled four-polytope with facet-plus-cusp sum fifteen is isometric to P_4. A right-angled five-polytope with sixteen facets is isometric to P_5. Consequently*

$$C \geq 11a_4 - (n-3)a_3, \qquad D \geq 17a_5 - (n-4)a_4.$$

Our published inputs are the standard truncation and Dehn–Sommerville framework described in [2], the lower bound eighty-two on eight-dimensional facet counts recorded there, and the four-dimensional bounds $F \geq 10$, $F + c \geq 15$ in [1, Section 4, equations (7)–(8)]. Compact right-angled polytopes have dimension at most four [1], so the noncompact eight-dimensional bound applies to every eight-face here. Equality classification is proved below; it is not assumed from published attainment of the bounds.

2 The weighted two-face calculation

Put $\lambda = 2123/1280$. For $n = 12$, truncation at the cusps gives

$$f = a_6 - 5a_7 + 14a_8 - 28a_9 + 42a_{10} - 42a_{11} - 462c,$$
$$a_2 = 15a_6 - 74a_7 + 206a_8 - 411a_9 + 616a_{10} - 616a_{11} - 2684c.$$

These rows follow directly by setting $b_0 = f + 2048c$, $b_{11} = a_{11} + c$ and $b_k = a_k + 2^{11-k}\binom{11}{k}c$ for $1 \le k \le 10$, and equating the coefficients of $H(t)$ and $t^{12}H(1/t)$ for

$$H(t) = (t-1)^{12} + \sum_{k=0}^{11} b_k (t-1)^k.$$

Eliminating gives

$$66f + 11264\lambda c - \frac{22}{5}a_2 = \frac{-22a_7 + 88a_8 - 198a_9 + 308a_{10} - 308a_{11}}{5} < 0. \tag{1}$$

Indeed $5a_7 \ge 82a_8$ and $3a_9 \ge 11a_{10}$, so in the numerator the first pair of terms is at most $-1364a_8/5$ and the second at most $-418a_{10}$.

A polygon with u finite and v ideal vertices satisfies $u + v \ge 3$ and $u + 2v \ge 5$, the latter by positive hyperbolic area. Summing these two inequalities and using (1) yields the two vertex-ratio bounds in dimension twelve. A counted triangle has weighted size $1 + 2\lambda$; every other polygon has size at least $3 + \lambda$. Hence

$$(3+\lambda)a_2 - (2-\lambda)T \le 66f + 11264\lambda c < \frac{22}{5}a_2,$$

which gives $T > 331a_2/437$.

For every noncounted polygon,

$$u + \lambda v - \frac{22}{5} \ge \frac{331}{3840}u.$$

It suffices to check the minimal pairs $(0,3), (2,2), (3,1), (5,0)$; every other permissible non-counted pair dominates one of them. A counted triangle has deficit $106/1280$. Summing gives $331(66f - T) < 318T$, and therefore $T > 1986f/59$.

For dimension eleven the exact coefficient rows are

$$f = 2a_6 - 12a_7 + 42a_8 - 112a_9 + 252a_{10} - 252c - 504,$$
$$a_2 = 25a_6 - 148a_7 + 515a_8 - 1370a_9 + 3080a_{10} - 1220c - 6160,$$
$$55f + 5120\lambda c - \frac{22}{5}a_2 = -\frac{44}{5}a_7 + 44a_8 - 132a_9 + 308a_{10} - 616 < 0.$$

Here $4a_7 \ge 82a_8$ and $2a_9 \ge 11a_{10}$ prove the strict sign. The same polygon argument gives the stated eleven-dimensional ratios and triangle densities.

3 Actual three- and four-faces

At a finite vertex let $q^2 = -1$ and let $n_1, \dots, n_n$ be its orthonormal outward normals. Form a graph G_q on the edge directions, joining i, j when their two-face is counted by T. Its opposite unit wall is uniquely

$$m_{ij} = q - n_i - n_j. \tag{2}$$

Its ideal endpoints are $q - n_i, q - n_j$.

Lemma 3. *A triangle in G_q supplies an actual three-face isometric to P_3.*

Proof. For the triangle i, j, k, the opposite point is $q' = 2q - n_i - n_j - n_k$, of square -1. For any other wall write $r = aq - \sum c_l n_l$, where $a > 0$ and $c_l \geq 0$. Acuteness with the three normals (2) gives $c_i + c_j, c_i + c_k, c_j + c_k \leq a$. Thus $r \cdot q' = -2a + c_i + c_j + c_k \leq -a/2 < 0$. The selected three m normals and the n_l outside the triple vanish there and are orthonormal. The six walls in the resulting three-space bound the convex hull of q, q' and the three ideal endpoints. All five points are feasible, so convexity identifies the whole face. □

Every right-angled three-polytope has at least six facets, with equality only for P_3. To prove this, write p, s, F for finite vertices, cusps and facets. Euler and vertex degrees give $p + 2s = 2F - 4$; summed polygon areas give $p + 6s \geq 20$. Together these exclude $F \leq 5$. If $F = 6$, they give $s \geq 3$. The polygon-corner sum is $24 - 2s$, so at least $2s$ of the six facets are triangular, forcing $s \leq 3$. Thus $s = 3, p = 2$ and all six polygons are counted triangles. Lemma 3 identifies the polytope.

Double counting now gives

$$\sum_q e(G_q) = T, \qquad \sum_q t(G_q) = 2B.$$

In dimension twelve, three-face Euler and the six-facet gap give

$$10a_2 = 2a_3 + 110f + 28160c, \qquad B \geq 7a_3 - 10a_2.$$

Since (1) gives $a_2 > 15f + 4246c$, we obtain

$$B > 7590c - 10f > \frac{2515}{59} f.$$

A four-clique-free graph on twelve vertices has at most sixty-four triangles. One elementary proof maximizes the weighted triangle sum on the vertex-weight simplex with minimum support: two nonadjacent supported vertices can be merged, so the support is a clique of size at most three. Its product is at most $1/27$; uniform weights give $12^3/27 = 64$. Thus some G_q has a four-clique.

For such a four-set S, its four n_i and six m_{ij} define the projective region $y_i \geq 0$, $y_i + y_j \leq t$, with $x = tq - \sum_{i \in S} y_i n_i$. Its vertices are q, the four points $2q - \sum_{i \in A} n_i$ for three-subsets $A \subset S$, four ideal points $q - n_i$, and the ideal point $2q - \sum_{i \in S} n_i$. The last is $m_{ij} + m_{kl}$ for a partition of S, and has nonpositive product with every other wall by acuteness. The remaining points are feasible by Lemma 3 and the original triangles. Their finite-volume convex hull is therefore the entire four-face: it is P_4.

Every four-polytope has at least ten facets. Indeed the preceding three-dimensional inequalities give $F + s \geq 9$ for each three-facet; its distinct neighbors and one distinct cusp-opposite per cusp force at least $1 + F + s$ ambient facets. At equality every facet has $F = 6, s = 3$, so the preceding four-clique argument identifies the whole polytope as P_4. Every other four-face has at least eleven facets, proving $C \geq 11a_4 - (n - 3)a_3$.

Deleting an edge of a four-clique until none remains takes at most $k_4(G)$ deletions. Each loses at most ten triangles on twelve vertices, so $t(G) \leq 64 + 10k_4(G)$. As $\sum_q k_4(G_q) = 5C$, the lower bound on B gives $C > 627f/1475$. In dimension eleven the corresponding identities are

$$a_3 = \frac{9}{2} a_2 - \frac{165}{4} f - 5760c, \qquad B > 3105c - \frac{15}{2} f > \frac{7545}{236} f.$$

The triangle bound is $11^3/27$, with integer part forty-nine; edge deletion loses at most nine triangles per edge. This yields $C > 1763f/5310$ and completes Theorem 1.

4 Minimum-sum four-dimensional rigidity

Distinct outward unit normals have product zero or at most -1. For if $-1 < u \cdot v < 0$, their positive definite Gram matrix has nonnegative inverse. Projecting a strict timelike center onto their orthogonal complement preserves all other strict wall inequalities by acuteness and gives an actual nonright ridge, a contradiction.

Lemma 4. *Two counted triangular facets sharing their ideal–ideal edge in a right-angled three-polytope force the whole polytope to be P_3.*

Proof. Normalize the common null endpoints by $u \cdot v = -1$ and let the orthogonal facet normals be n_1, n_2. Their finite apices are $u + v - n_2$ and $su + s^{-1}v - n_1$, $s > 0$. The other four walls have normals $u - n_2, v - n_2, su - n_1, s^{-1}v - n_1$. Two cross products are $-s, -1/s$, so the normal-product gap forces $s = 1$. The third cusp $u + v - n_1 - n_2 = (u - n_1) + (v - n_2)$ is feasible by acuteness. All five vertices of the six-wall bipyramid are therefore feasible, proving the claim by convexity. □

Suppose a four-polytope has $F + c = 15$. The published $F \geq 10$ gives $c \leq 5$. A three-facet with r polygon facets and s cusps satisfies $r + 2s \geq 12$. Summing gives degree sum at least $12F - 12c$ in the facet graph, since every cusp lies in six facets. This graph has no five-clique: mutually adjacent normals would be five orthogonal spacelike vectors in positive index four. Its degree sum is at most $3F^2/4$ by the elementary weighted clique bound. The lower bound exceeds this for $c = 0, 1, 2$. For $c = 3, F = 12$, equality forces all degrees nine: uniform weights maximize the edge quadratic in the interior of the simplex, so its first derivatives are equal. But equality also gives $r + 2s = 12$ in every facet, making every degree even, a contradiction.

For $c = 4, F = 11$, the neighbor/cusp-opposite injection gives $r + s \leq 10$. Together with $r + 2s \geq 12$ this gives $s \geq 2$. Values $s \geq 4$ would give $r \leq 6$, contrary to the six-facet equality classification. Since $\sum s = 24$, nine facets have two cusps and two have three. A two-cusp facet has $r = 8$ and zero total polygon excess (sides) + (ideal vertices) $- 5$. If it has a triangle, the polygon across its ideal–ideal edge is also a counted triangle; Lemma 4 contradicts its eight facets. Its thirty-two polygon corners thus make all eight polygons quadrilaterals. A three-cusp facet has six or seven polygons, at least four triangular. Each would need a different adjacent facet with a triangular polygon, whereas only the other three-cusp facet can do so. This excludes $c = 4$. Finally $c = 5$ gives $F = 10$ and hence P_4.

5 The sixteen-facet five-polytope

The injection and $F + c \geq 15$ force at least sixteen facets in dimension five. At equality every four-facet has sum fifteen and is P_4. Pick a finite vertex q with orthonormal normals $n_1, \ldots, n_5$. All ten m_{ij} of (2) are actual walls, giving fifteen walls. For each four-set S the forced P_4 has cusp $v_S = 2q - \sum_{i \in S} n_i$. Exactly seven known walls meet it: the omitted n_k and six m_{ij}. Its rectangular four-box link needs one more wall, the remaining sixteenth wall r, opposite n_k. Thus $r + n_k = L_k v_S$, $L_k > 0$. Comparing the q coefficient and then an n_k coefficient makes every $L_k = 1$. Consequently

$$r = 2q - \sum_{i=1}^{5} n_i.$$

The sixteen walls define $y_i \geq 0$, $y_i + y_j \leq 1$, $\sum y_i \leq 2$. Its exact vertex enumeration gives the origin; five ideal axes; ten finite half-triples; five ideal half-four-sets; and five finite points with one coordinate $2/3$ and the others $1/3$. There are sixteen finite and ten ideal vertices. Every facet is P_4. Thus the polytope is P_5, proving Theorem 2.

6 Complete five-corners and a sharp twenty-facet gap

Call a finite corner complete when its triangle graph is complete.

Theorem 5. *A right-angled five-polytope with a complete corner is either P_5 or has at least twenty facets. In the latter case, if L, V count its complete corners and finite vertices, then $11L \le V$. Both bounds are sharp for the reflected double of P_5 across the wall $\rho = 2q - \sum_{i=1}^{5} n_i$: it has twenty facets, twenty-two finite vertices, fifteen cusps and two complete corners.*

Proof. The five n_i and ten m_{ij} at a complete corner supply fifteen actual walls. The canonical sixteen-wall P_5 seed is contained in the polytope even if ρ is absent. All but its five extra finite vertices are feasible by the three- and four-face constructions. For the remaining points $z_i = 3q - 2n_i - \sum_{j\neq i} n_j$, write an additional wall as $r = aq - \sum c_j n_j$. We need $2c_i + \sum_{j\neq i} c_j \le 3a$, knowing $c_j \ge 0$ and $c_i + c_j \le a$. If $c_i \ge a/2$, summing its four pair bounds gives at most $4a - 2c_i \le 3a$. Otherwise split the other four indices into two pairs to get at most $2c_i + 2a \le 3a$. Convexity proves containment.

At each cusp $v_i = 2q - \sum_{j\neq i} n_j$, the six m_{jk} form three opposite pairs; with n_i these are seven of its eight walls. The remaining wall is $r_i = \lambda_i v_i - n_i$, $\lambda_i > 0$. Since $r_i \cdot m_{ij} = 1 - \lambda_i$, the normal-product gap forces $\lambda_i = 1$ or $\lambda_i \ge 2$. For different i, j,

$$r_i \cdot r_j = 1 - (\lambda_i - 1)(\lambda_j - 1).$$

If some $\lambda_i = 1$, product one forces all these walls to coincide; coefficient comparison makes every $\lambda_j = 1$ and the common wall is ρ. The sixteen actual walls and seed containment give $P = P_5$. Otherwise the five r_i are distinct and have q coefficient at least four, so none is among the first fifteen. There are at least twenty walls.

Suppose now $P \neq P_5$. Each complete corner has ten distinct opposite finite vertices across its P_3 faces. Such a recipient is not complete: its edge on the two untouched normals would otherwise force ρ. In fact a recipient admits at most one complete predecessor. To see this, label a receiving triple $S = \{1, 2, 3\}$ at a recipient p. The complete opposite is $p_S = 2p - n_1 - n_2 - n_3$, with incident normals $m_{23}, m_{13}, m_{12}, n_4, n_5$. Its complete graph forces all cross walls

$$p_S - m_{ij} - n_l = p - n_k - n_l, \qquad \{i, j, k\} = S, \quad l \notin S.$$

These supply actual counted triangles at p: their null endpoints are sums of known outward normals and hence feasible, and the selected three walls bound their entire triangle. Therefore the receiving graph is exactly K_5 minus the untouched edge. Its unique missing edge fixes S and the predecessor. Counting gives $10L \le V - L$. The exact reflected-double calculation proves sharpness. □

If $K = \sum_q k_5(G_q)$ in ambient dimension n, summing over five-faces gives the further universal inequality

$$11K \le \binom{n}{5} f + 160D. \tag{3}$$

Indeed a canonical five-face has sixteen complete finite corners, whereas every other five-face satisfies Theorem 5.

7 Propagation cannot close in dimension at least nine

Proposition 6. *For $n \ge 9$, if a finite-volume right-angled n-polytope has a complete finite corner, some five-face with a complete corner is not P_5.*

Proof. Across any P_3 opposite the vector $h = 3q - \sum_{i=1}^{n} n_i$ is unchanged: the three new normals are the three m_{ij}, and the others stay fixed. Thus $h^2 = n - 9$ and $h \cdot q = -3$ throughout each opposition component. If every complete five-corner lies in P_5, a complete n-corner propagates to every P_3 opposite. The three new directions and their cross edges are forced by the old complete graph; every pair of untouched directions is forced by its canonical five-face. The resulting finite component is therefore regular of degree $d = \binom{n}{3}$, entirely consisting of complete corners. At each of its vertices,

$$\frac{1}{d} \sum_{|S|=3} q_S = \left(2 - \frac{9}{n}\right) q + \frac{3}{n} h.$$

Summing over its N vertices yields $(9 - n) \sum q = 3Nh$. For $n = 9$ this contradicts $h \neq 0$. For $n > 9$ it equates a timelike vector with a spacelike one. □

This is a propagation obstruction, not yet a quantitative density contradiction. It does not by itself lower a dimension ceiling.

8 A universal opposition moment bound in dimension twelve

Proposition 7. *Every finite-volume right-angled twelve-polytope satisfies*

$$2B \leq \frac{1267552}{6875} f < 184.372 f.$$

Proof. On each opposition component, $h^2 = 3$. Choose a future null vector $z \perp h$, put $E(q) = -z \cdot q > 0$ and $p_i = -z \cdot n_i / E(q)$. Then $\sum p_i = 3$, $\sum p_i^2 = 1$. For each of the 220 formal triples, whether or not it supplies an actual face,

$$r_S = \frac{E(q_S)}{E(q)} = 2 - \sum_{i \in S} p_i.$$

Set $x_i = p_i - 1/4$, $\mu = 5/4$ and $d_S = -\sum_{i \in S} x_i$. Thus $r_S = \mu + d_S$, $\sum x_i = 0$, $\sum x_i^2 = 1/4$. Cauchy–Schwarz after subtracting the mean of the three-set indicator gives $|d_S| \leq 3/4$, hence $1/2 \leq r_S \leq 2$.

Uniform averages over the triples satisfy

$$\operatorname{av} d = 0, \qquad \operatorname{av} d^2 = \frac{9}{176},$$
$$\operatorname{av} d^3 = -\frac{27}{220} \sum x_i^3, \qquad \operatorname{av} d^4 = \frac{3}{440} - \frac{3}{220} \sum x_i^4 \leq \frac{3}{440}.$$

Expand the powers using the subset inclusion probabilities

$$1/4, \qquad 1/22, \qquad 1/220.$$

Also

$$|\sum x_i^3| \leq (\sum x_i^2)^{3/2} = 1/8.$$

The exact reciprocal expansion

$$\frac{1}{\mu + d} = \frac{1}{\mu} - \frac{d}{\mu^2} + \frac{d^2}{\mu^3} - \frac{d^3}{\mu^4} + \frac{d^4}{\mu^4(\mu + d)}$$

therefore gives

$$\operatorname{av} \frac{1}{r_S} \leq \frac{4}{5} + \frac{64}{125}\frac{9}{176} + \frac{256}{625}\frac{27}{1760} + \frac{512}{625}\frac{3}{440} = \frac{28808}{34375}.$$

With $g(q) = 1/E(q)$, the sum of $g(q_S)/g(q)$ over actual neighbors is at most 220 times this value. Summing over vertices, each undirected opposition edge contributes $g(v)/g(u) + g(u)/g(v) \geq 2$. There are B edges, which proves the proposition. □

For linear objectives involving only clique counts on twelve vertices, all corner graph vectors can be replaced by convex mixtures of the 77 complete multipartite vectors. Indeed, partition any maximizing graph into maximal independent classes with identical external neighborhoods. For two nonadjacent classes A, B, no clique meets both, so the objective is a constant plus $|A|d_A + |B|d_B$. Replace the lower-contribution class by clones of the other. Other classes remain intact; the class count decreases and the objective does not decrease, even for signed objective coefficients. Iteration gives a complete multipartite graph. Equality of all linear support functions proves the convex-hull statement. This does not identify an individual geometric corner graph as multipartite.

9 Reproduction, diagnostic and verification scope

The standard-library exact checkers include

```
scripts/check_right_angled_mixed_weighted_triangles_research_20261018.py
scripts/check_right_angled_mixed_low_facet_rigidity_research_20261018.py.
```

The complete-corner and opposition-moment additions are checked by

```
scripts/check_right_angled_mixed_complete_five_corner_research_20261018.py
scripts/check_right_angled_mixed_opposition_moments_research_20261018.py.
```

They recompute the Dehn–Sommerville equations by rational elimination, the numerical inequalities, the twenty and 210 active-wall choices for P_3, P_4, and all 4368 five-wall choices for P_5. The additional checks reconstruct the twenty-facet double and its corner graphs, and verify the exact moment coefficients and constants. Generation, fresh replays and independent written/source reviews pass. The scripts verify coefficient and model claims; the universal geometric arguments are reviewed written proofs, not encoded by those enumerations.

The separate exact diagnostic

```
scripts/check_right_angled_mixed_incidence_feasibility_research_20261018.py
```

constructs a simultaneous numerical solution of the earlier dimension-twelve aggregate constraints, including canonical incidences and integer-scaled Macaulay conditions. Its original assignment violates the new moment cap. The moment packet supplies a modified assignment satisfying the new scalar cap and packing inequality as well; its complete-corner saturation still conflicts with the propagation argument above. No quantitative inequality yet excludes every nearby assignment. None of these assignments supplies hyperplanes, a boundary complex, consistent cusp widths or geometric gluing.

Run `scripts/check_mixed_corner_package_20261018.py--build` to replay the five small checkers sequentially and compile this paper. The runner uses the shared exclusive computation lock and resource gates, retains a compact audit and the PDF, and removes TeX intermediates.

No new example or dimension ceiling is claimed. The new geometric theorems here await Lean verification. The repository's existing Lean results do not certify these arguments merely because their final inequalities are rational.

References

[1] L. Potyagailo and E. Vinberg, On right-angled reflection groups in hyperbolic spaces, *Comment. Math. Helv.* 80 (2005), 63–73. `https://ems.press/content/serial-article-files/43032`.

[2] S. Alexandrov, On ideal vertices of right-angled hyperbolic polyhedra, 2023, Sections 2–6. `https://arxiv.org/html/2303.09533v1`.